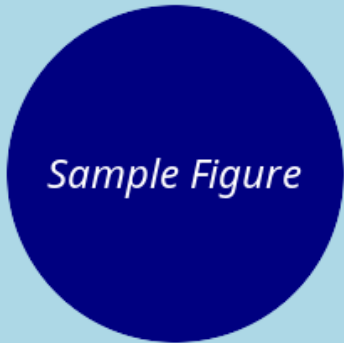


## Graphical Abstract

**MyST Markdown with the Elsevier CAS and elsarticle Classes: Feature Reference and Template Examples**

A sample figure consisting of a dark blue circle centered on a light blue square background. The text "Sample Figure" is written in white, italicized font inside the circle.

*Sample Figure*

## Highlights

### **MyST Markdown with the Elsevier CAS and elsarticle Classes: Feature Reference and Template Examples**

- MyST Markdown enables reproducible scientific writing
- Seamless export to multiple journal formats
- Rich mathematical and scientific notation support

# MyST Markdown with the Elsevier CAS and elsarticle Classes: Feature Reference and Template Examples

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## Abstract

This article demonstrates MyST Markdown integration with Elsevier’s CAS and elsarticle classes, covering typography, math, cross-references, admonitions, proofs, tables, figures, and code blocks.

*Keywords:* MyST Markdown, Elsevier, LaTeX, CAS Template, Scientific Publishing, Reproducible Research

*JEL:* D14; C61; G11

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## Summary

One Markdown source produces both Elsevier layouts and a website.

*Beware of bugs in the above code; I have only proved it correct, not tried it.*

— Donald E. Knuth

## 1. Introduction

This document demonstrates the full integration of MyST Markdown [1] with Elsevier’s CAS templates. MyST provides a powerful authoring experience while maintaining compatibility with traditional LaTeX journal requirements [2]. The template renders to PDF for camera-ready submission and to HTML for the project website; author identifiers (ORCID) and persistent links (DOI) propagate through both pipelines.

### 1.1. Background

Scientific publishing has traditionally relied on LaTeX for high-quality typesetting [3]. However, the learning curve and complexity of LaTeX can be a barrier for authors who write outside it day to day. MyST Markdown bridges this gap by providing:

1. A familiar Markdown syntax based on CommonMark [4]

2. Rich scientific features (equations, citations, cross-references)
3. Export to multiple formats including PDF via LaTeX

Reproducible research workflows have become increasingly important, with tools like Jupyter Notebooks [5] enabling literate programming approaches.

### Note

**About this document:** This article is both a feature reference and a test file for the Elsevier MyST template. Every feature shown here should render correctly in the web preview and in all six PDF exports, under CAS and elsarticle alike.

## 2. Typography Features

This section demonstrates MyST Markdown typography features and how they render in the PDF output.

### 2.1. Inline Formatting

Standard inline formatting includes **bold text**, *italic text*, and `inline code`. You can also use ~~struckthrough text~~ and underlined text for special emphasis.

For chemical formulas, use subscripts: H<sub>2</sub>O, CO<sub>2</sub>, C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>. For ordinals and exponents, use superscripts: the 4<sup>th</sup> of July, 1<sup>st</sup> place,  $x^2 + y^2 = r^2$ .

## 2.2. Line Breaks

The world’s shortest poem demonstrates line breaks:

Fleas

Adam

Had ’em.

—Strickland Gillilan

## 2.3. Quotations

Block quotes are useful for highlighting important passages:

[Figure 1 about here.]

[Figure 2 about here.]

[Figure 3 about here.]

## 2.4. Definition Lists

MyST supports definition lists for glossaries or term explanations:

**MyST** Markedly Structured Text, a markdown flavor for scientific writing

**LaTeX** A document preparation system for high-quality typesetting

**jtex** A Jinja-based templating system for LaTeX documents

**CAS** Complex Article Service, the Elsevier workflow the cas-sc and cas-dc classes target

## 2.5. Footnotes

MyST supports footnotes<sup>1</sup> which are automatically numbered and placed at the end of the document. You can have multiple footnotes<sup>2</sup> throughout your text.

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<sup>1</sup>This is a footnote demonstrating the feature. Footnotes can contain **formatted text** and even `code`.

<sup>2</sup>Another footnote with additional information. Footnotes are essential for academic writing.

## 2.6. Task Lists

Task lists can track progress (rendered as bullet points in LaTeX):

- Create template structure
- Add typography examples
- Add math demonstrations
- Add proof environments
- Submit to journal
- Celebrate publication!

## 3. Mathematical Content

The templates support full LaTeX math with custom macros defined in frontmatter.

### 3.1. Inline Mathematics

Inline math like  $E = mc^2$  or  $\mathbb{E}[X] = \mu$  sets in running text without extra markup. Using our custom macros: for  $x \in \mathbb{R}$ , we have  $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$  and  $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ .

### 3.2. Display Equations

The quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

Maxwell’s equations in differential form:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned} \quad (2)$$

The Bellman equation for dynamic programming, where the policy is the maximizing action  $a^*(s) = \text{argmax}_a \{R(s, a) + \gamma \sum_{s'} P(s'|s, a)V(s')\}$ :

$$V(s) = \max_a \left\{ R(s, a) + \gamma \sum_{s'} P(s'|s, a) V(s') \right\} \quad (3)$$

Equations can be cross-referenced: see (1) for the quadratic formula and (2) for Maxwell's equations.

## 4. Proofs and Theorems

MyST supports formal mathematical environments using proof directives. These are essential for mathematical and theoretical papers.

### 4.1. Definitions

**Definition 4.1** (Convergent Sequence). *A sequence  $(a_n)$  in  $\mathbb{R}$  is said to be convergent if there exists a number  $L \in \mathbb{R}$  such that for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that for all  $n > N$ :*

$$|a_n - L| < \varepsilon \quad (4)$$

We write  $\lim_{n \rightarrow \infty} a_n = L$ .

**Definition 4.2** (Continuous Function). *A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is continuous at a point  $c$  if for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that:*

$$|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon \quad (5)$$

### 4.2. Theorems

**Theorem 4.1** (Squeeze Theorem). *Let  $(a_n)$ ,  $(b_n)$ , and  $(c_n)$  be sequences such that  $a_n \leq b_n \leq c_n$  for all  $n \geq N_0$ .*

*If*

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L \quad (6)$$

*then  $\lim_{n \rightarrow \infty} b_n = L$ .*

**Theorem 4.2** (Fundamental Theorem of Calculus). *Let  $f$  be continuous on  $[a, b]$  and let  $F$  be defined by  $F(x) = \int_a^x f(t) dt$ . Then:*

1.  $F$  is continuous on  $[a, b]$
2.  $F$  is differentiable on  $(a, b)$  and  $F'(x) = f(x)$
3.  $\int_a^b f(x) dx = F(b) - F(a)$

### 4.3. Proofs

*Proof. Proof of Theorem 4.1.* Let  $\varepsilon > 0$  be given. Since  $a_n \rightarrow L$  and  $c_n \rightarrow L$ , there exist  $N_1, N_2 \in \mathbb{N}$  such that:

- $|a_n - L| < \varepsilon$  for all  $n > N_1$
- $|c_n - L| < \varepsilon$  for all  $n > N_2$

Let  $N = \max\{N_0, N_1, N_2\}$ . For  $n > N$ , we have:

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon \quad (7)$$

Thus  $|b_n - L| < \varepsilon$ , completing the proof. □

### 4.4. Lemmas and Corollaries

**Lemma 4.3** (Triangle Inequality). *For all  $x, y \in \mathbb{R}$ , we have  $|x + y| \leq |x| + |y|$ .*

**Corollary 4.3.1.** *If  $(b_n)$  is squeezed between two sequences converging to zero, then  $b_n \rightarrow 0$ .*

### 4.5. Remarks and Examples

**Remark 4.1.** *The proof directives (Definition 4.1, Theorem 4.1) are automatically numbered and can be cross-referenced throughout the document.*

**Example 4.1** (Convergent Sequence). *The sequence  $a_n = \frac{1}{n}$  converges to 0. For any  $\varepsilon > 0$ , choose  $N > \frac{1}{\varepsilon}$ . Then for  $n > N$ :*

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \frac{1}{N} < \varepsilon \quad (8)$$

## 5. Admonitions

Admonitions (callouts) are useful for highlighting important information. MyST supports many types:

### Note

This is a note admonition. Use it to highlight supplementary information that readers should be aware of.

### Warning

Use a warning to alert readers about pitfalls or caveats in your methodology.

### Tip

Tips can provide helpful suggestions for readers applying your methods.

### Important

Important information that readers must not miss should go in this type of admonition.

### Hint

Hints can be hidden in dropdowns! Click to reveal.

This hint contains the secret formula:  $e^{i\pi} + 1 = 0$

### Caution

Caution: Proceed carefully when implementing this algorithm with large datasets.

### Attention

Attention: New methodology introduced in this section differs from previous work.

### Danger

**Danger:** Do not run this code in production without proper testing!

### Error

Error: This approach will fail if the input matrix is singular.

### See Also

For more information on MyST Markdown features, visit the [MyST documentation](#).

## 6. Code Blocks

Code blocks with syntax highlighting are supported:

[Figure 4 about here.]

As shown in Program B.1, code can be captioned, numbered, and cross-referenced.

Multiple languages are supported:

[Figure 5 about here.]

## 7. Algorithms

For algorithm pseudocode, use raw LaTeX blocks with the `algorithm` and `algorithmic` environments:

---

### Algorithm 1 Binary Search

---

**Require:** Sorted array  $A[1..n]$ , target value  $x$

**Ensure:** Index  $i$  such that  $A[i] = x$ , or  $-1$  if not found

```
1:  $low \leftarrow 1$ 
2:  $high \leftarrow n$ 
3: while  $low \leq high$  do
4:    $mid \leftarrow \lfloor (low + high)/2 \rfloor$ 
5:   if  $A[mid] = x$  then
6:     return  $mid$ 
7:   else if  $A[mid] < x$  then
8:      $low \leftarrow mid + 1$ 
9:   else
10:     $high \leftarrow mid - 1$ 
11:  end if
12: end while
13: return  $-1$ 
```

---

The algorithm environment provides a float with caption and label for cross-referencing.

## 8. Tables

### 8.1. Markdown Tables

MyST tables convert cleanly to LaTeX:

[Table 1 about here.]

Results are summarized in Table B.1.

### 8.2. List Tables

List tables provide an alternative syntax useful for complex content:

[Table 2 about here.]

### 8.3. CSV Tables

[Table 3 about here.]

### 8.4. Raw LaTeX Tables

For complex tables requiring advanced features, use raw LaTeX blocks. The CAS templates include `booktabs`, `multirow`, `makecell`, `array`, and `dcolum` packages:

[Table 4 about here.]

## 9. Figures and Images

[Figure 6 about here.]

As shown in Figure B.3, the template properly handles figure placement and captions.

## 10. Cross-References Summary

This document demonstrates various cross-reference capabilities:

- **Equations:** (1), (2), (3)
- **Figures:** Figure B.3
- **Tables:** Table B.1, Table B.2, Table B.3
- **Code:** Program B.1, Program B.2

- **Algorithms:** Algorithm 1 (via raw LaTeX `\ref{alg:binary-search}`)
- **Theorems:** Theorem 4.1, Theorem 4.2
- **Definitions:** Definition 4.1, Definition 4.2
- **Lemmas:** Lemma 4.3
- **Corollaries:** Corollary 4.3.1
- **Examples:** Example 4.1
- **Sections:** Introduction, Typography

## 11. Discussion

This approach enables researchers to write in MyST Markdown while producing publication-ready documents that meet Elsevier's submission requirements.

### 11.1. Advantages

1. **Reproducibility:** Source files are plain text and version-controllable
2. **Flexibility:** Single source exports to HTML, PDF, and other formats
3. **Modern tooling:** Integration with Jupyter, VS Code, and other tools
4. **Rich features:** Full LaTeX math, cross-references, and citations

### 11.2. Limitations

#### Warning

Some MyST features render only in HTML output. For PDF, consider using LaTeX-compatible alternatives.

## 12. Conclusion

The Elsevier CAS MyST template provides a modern workflow for scientific writing while maintaining compatibility with traditional journal submission systems. This document has demonstrated:

- Typography: formatting, footnotes, definition lists
- Mathematics: inline, display, custom macros
- Formal environments: definitions, theorems, proofs, examples
- All admonition types
- Multiple table formats
- Code blocks with syntax highlighting
- Figures with cross-referencing
- Citations and bibliography

For questions or contributions, please visit the template repository.

## Appendix A. Supplementary Methods

This appendix provides additional methodological details that support the main text.

### Appendix A.1. Data Processing

The data was processed using standard procedures as described in the literature.

This paragraph cites [6], a work no other section cites. It is here to keep the appendix on the bibliography-harvesting path: while this file was pulled in through `parts.appendix`, a citation made here reached the `.tex` and never the `.bib`, and no build step reported anything.

## Appendix B. Additional Tables

[Table 5 about here.]

## Declarations

The authors declare no competing interests and did not use generative AI or AI-assisted technologies to prepare this work. This statement exercises the declaration part, which the template prints under the heading Declarations between the CRediT statement and the references.

## Data Availability

No data were used for the research described in the article.

## References

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Computational Workflows, in: Positioning and Power in Academic Publishing: Players, Agents and Agendas, IOS Press, 2016, pp. 87–90. [doi:10.3233/978-1-61499-649-1-87](https://doi.org/10.3233/978-1-61499-649-1-87).

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Alan E. Lujan develops open-source computational tools for economics at Johns Hopkins University and Econ-ARK.

Christopher D. Carroll is a professor of economics at Johns Hopkins University and leads the Econ-ARK initiative for reproducible heterogeneous-agent macroeconomics.

## List of Figures

We know what we are, but know not what we may be.

William Shakespeare, Hamlet

The important thing is not to stop questioning. Curiosity has its own reason for existing.

In the middle of difficulty lies opportunity.

Albert Einstein

```

import numpy as np
from typing import Tuple

def quadratic_formula(a: float, b: float, c: float) -> Tuple[float, float]:
    """Solve  $ax^2 + bx + c = 0$  using the quadratic formula."""
    discriminant = b**2 - 4*a*c
    if discriminant < 0:
        raise ValueError("No real solutions")
    x1 = ( -b + np.sqrt(discriminant)) / (2*a)
    x2 = ( -b - np.sqrt(discriminant)) / (2*a)
    return x1, x2

# Example usage
roots = quadratic_formula(1, -5, 6)
print(f"Roots: {roots}") # Output: (3.0, 2.0)

```

Figure B.1: Example Python implementation of the quadratic formula

```
function quadratic_formula(a, b, c)
    discriminant = b^2 - 4*a*c
    x1 = ( -b + sqrt(discriminant)) / (2*a)
    x2 = ( -b - sqrt(discriminant)) / (2*a)
    return x1, x2
end
```

Figure B.2: Julia implementation

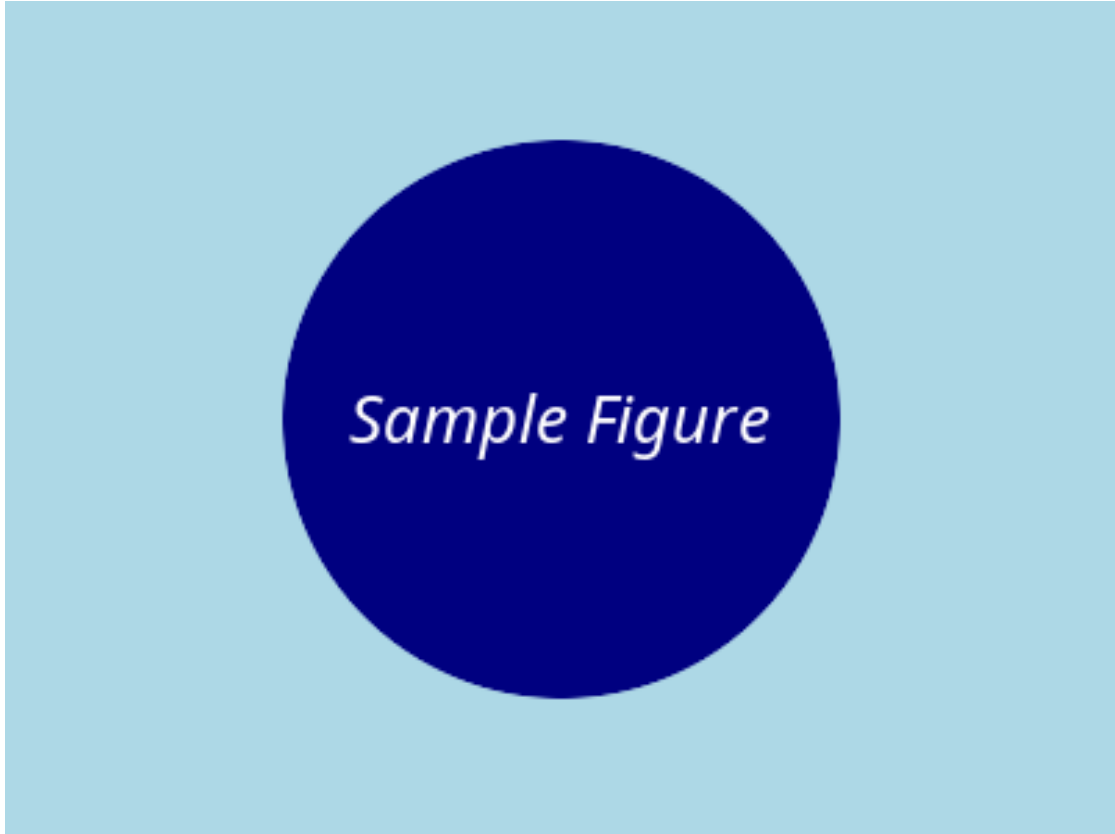


Figure B.3: A sample figure demonstrating image support in the template. This figure shows a placeholder image that would typically contain research results or visualizations. Figures are automatically numbered and can be cross-referenced.

**List of Tables**

B.4 Table features: multi-column headers, aligned decimals, and stacked cells . . . . . 19

Table B.1: Comparison of numerical methods

| Method   | Accuracy     | Speed  | Memory |
|----------|--------------|--------|--------|
| Baseline | 85.2%        | Fast   | Low    |
| Proposed | <b>92.1%</b> | Medium | Medium |
| Ensemble | 94.3%        | Slow   | High   |
| Oracle   | 98.5%        | N/A    | N/A    |

Table B.2: Dataset characteristics

| Dataset  | Training Samples | Test Samples | Features | Classes |
|----------|------------------|--------------|----------|---------|
| MNIST    | 60,000           | 10,000       | 784      | 10      |
| CIFAR-10 | 50,000           | 10,000       | 3,072    | 10      |
| ImageNet | 1,281,167        | 50,000       | 150,528  | 1,000   |

Table B.3: Experimental results

| Model               | Accuracy | Precision | Recall | F1-Score |
|---------------------|----------|-----------|--------|----------|
| Logistic Regression | 0.82     | 0.81      | 0.83   | 0.82     |
| Random Forest       | 0.89     | 0.88      | 0.90   | 0.89     |
| Neural Network      | 0.94     | 0.93      | 0.95   | 0.94     |
| Transformer         | 0.97     | 0.96      | 0.97   | 0.97     |

Table B.4: Table features: multi-column headers, aligned decimals, and stacked cells

| Category |                          | Measurements |       |       |
|----------|--------------------------|--------------|-------|-------|
| Type     | Description<br>(details) | Status       | Value | Count |
| Group A  | Item 1                   | Yes          | 12.34 | 100   |
|          | Item 2                   | Yes          | 5.67  | 250   |
|          | Item 3                   | -            | 89.01 | 50    |
| Group B  | Item 4                   | Yes          | 23.45 | 175   |
|          | Item 5                   | -            | 6.78  | 320   |

Features: *booktabs* rules, *multirow* row spanning, *makecell* line breaks, *dcolumn* decimal alignment, *array* column formatting.

Table B.5: Supplementary parameters

| Parameter | Value | Unit |
|-----------|-------|------|
| Alpha     | 0.05  | -    |
| Beta      | 1.23  | m/s  |
| Gamma     | 456   | kg   |